1. ZONE

Attributes: zone_id, zone_name, zone_headquarters

Candidate Keys: {zone_id}, {zone_name}

All FDs:

- `zone_id → zone_name`
- `zone_id → zone_headquarters`
- zone_name → `{zone_id, zone_headquaters}`

BCNF Check: All non-trivial FDs have a candidate key (zone_id or zone_name) on the left, so every determinant is a superkey and the relation is in BCNF.

✓ zone is in BCNF.

2. DIVISION

Attributes: division_id, zone_id, division_name, division_headquarters

Candidate Keys: {division_id}

All FDs:

- `division_id → zone_id`
- `division_id → division_name`
- `division_id → division_headquarters`

BCNF Check: Division_id is candidate key, so every FD has a superkey on the left and the relation is in BCNF.

✓ division is in BCNF.

3. DEPARTMENT

Attributes: dept_id, dept_name, description, head_officer_title

Candidate Keys: {dept_id}, {dept_name}

All FDs:

- `dept_id → dept_name`
- `dept_id → description`

- `dept_id → head_officer_title`
- `dept_name → {dept_id, description, head_officer_title}`

BCNF Check: All determinants (dept_id, dept_name) are candidate keys, so every FD has a superkey on the left and the relation is in BCNF.

✓ department is in BCNF.

4. STATION

Attributes: station_id, division_id, station_name, state, city, is_junction

Candidate Keys: {station_id}

All FDs:

- `station_id → division_id`
- `station_id → station_name`
- `station_id → state`
- `station_id → city`
- `station_id → is_junction`

BCNF Check: All FDs have station_id as determinant, which is the sole candidate key. No non-key attribute determines another.

✓ station is in BCNF.

5. ROUTE

Attributes: route_id, route_name, total_distance

Candidate Keys: {route_id}, {route_name}

All FDs:

- `route_id → route_name`
- `route_id → total_distance`
- `route_name → {route_id, total_distance}`

BCNF Check: All determinants (route_id, route_name) are candidate keys, so every FD has a superkey on the left and the relation is in BCNF.

✓ route is in BCNF.

6. TRAIN_CLASS

Attributes: class_id, class_name, fare_multiplier

Candidate Keys: {class_id}, {class_name}

All FDs:

- `class_id → class_name`
- `class_id → fare_multiplier`
- `class_name → {class_id, fare_multiplier}`

BCNF Check: All determinants (class_id, class_name) are candidate keys, so every FD has a superkey on the left and the TRAIN_CLASS relation is in BCNF.

✓ train_class is in BCNF.

7. REGISTERED_USER

Attributes: user_id, name, email, password, dob, address, phone

Candidate Keys: {user_id}, {email}, {phone}

All FDs:

- `user_id → name`
- `user_id → email`
- `user_id → password`
- `user_id → dob`
- `user_id → address`
- `user_id → phone`
- `email → {user_id, name, password, dob, address, phone}`
- `phone → {user_id, name, email, password, dob, address}`

BCNF Check: For this REGISTERED_USER relation, all three determinants (user_id, email, phone) are candidate keys, so every FD has a superkey on the left and the table is in BCNF.

✓ registered_user is in BCNF.

8. MAINTENANCE_SHED

Attributes: shed_id, division_id, shed_name, shed_type, capacity, inspection_flag

Candidate Keys: {shed_id}

All FDs:

- `shed_id → division_id`
- `shed_id → shed_name`
- `shed_id → shed_type`
- `shed_id → capacity`
- `shed_id → inspection_flag`

BCNF Check: All FDs have shed_id as determinant, which is the sole candidate key.

✓ maintenance_shed is in BCNF.

9. LOCOMOTIVE

Attributes: loco_id, loco_class, shed_id

Candidate Keys: {loco_id}

All FDs:

- `loco_id → loco_class`
- `loco_id → shed_id`

BCNF Check: All FDs have loco_id as determinant, which is the sole candidate key.

✓ locomotive is in BCNF.

10. TRAIN

Attributes: train_id, route_id, loco_id, train_name, train_status, total_coaches, class_id, base_fare,train_type

Candidate Keys: {train_id}, {loco_id}

All FDs:

- `train_id → route_id`
- `train_id → loco_id`
- `train_id → train_name`
- `train_id → train_status`
- `train_id → total_coaches`
- `train_id → class_id`
- `train_id → base_fare`
- `train_id → train_type`

- `loco_id → {train_id, route_id, train_name, train_status, train_coaches, class_id, base_fare, train_type}`

BCNF Check: All determinants in TRAIN (train_id, loco_id) are candidate keys, so every FD has a superkey on the left and the relation is in BCNF.

✓ train is in BCNF.

11. MAINTENANCE_RECORD

Attributes: maintenance_id, train_id, maintenance_date, maintenance_type, description, maintenance_status

Candidate Keys: {maintenance_id}

All FDs:

- `maintenance_id → train_id`
- `maintenance_id → maintenance_date`
- `maintenance_id → maintenance_type`
- `maintenance_id → description`
- `maintenance_id → maintenance_status`

BCNF Check: All FDs have maintenance_id as determinant, which is the sole candidate key.

✓ maintenance_record is in BCNF.

12. STAFF

Attributes: staff_id, station_id, name, role, phone, email, salary, dept_id

Candidate Keys: {staff_id}, {email}, {phone}

All FDs:

- `staff_id → station_id`
- `staff_id → name`
- `staff_id → role`
- `staff_id → phone`
- `staff_id → email`
- `staff_id → salary`
- `staff_id → dept_id`
- `email → {staff_id, station_id, name, role, phone, salary, dept_id}`
- `phone → {staff_id, station_id, name, role, email, salary, dept_id}`

BCNF Check: All determinants (staff_id, email, phone) are candidate keys, so every FD has a superkey on the left and the STAFF relation is in BCNF.

✓ staff is in BCNF.

13. MAINTENANCE_STAFF

Attributes: staff_id, maintenance_id

Candidate Keys: {staff_id, maintenance_id}

All FDs:

- `No non-trivial FDs (only key attributes)`

BCNF Check: A relation with only key attributes trivially satisfies BCNF — there are no non-trivial FDs to violate it.

✓ maintenance_staff is in BCNF.

14. TICKET_CHECKER

Attributes: staff_id, fine_log

Candidate Keys: {staff_id}

All FDs:

- `staff_id → fine_log`

BCNF Check: All FDs have staff_id as determinant, which is the sole candidate key.

✓ ticket_checker is in BCNF.

15. DRIVER

Attributes: staff_id, train_id, driving_license_no

Candidate Keys: {staff_id}, {driving_license_no}

All FDs:

- `staff_id → train_id`
- `staff_id → driving_license_no`

- `driving_license_no → {staff_id, train_id}`

BCNF Check: All determinants (staff_id, driving_license_no) are candidate keys, so every FD has a superkey on the left and the DRIVER relation is in BCNF.

No partial dependencies (neither candidate key is composite). No transitive dependency.

✓ driver is in BCNF.

16. GUARD

Attributes: staff_id, train_id, security_clearance

Candidate Keys: {staff_id}

All FDs:

- `staff_id → train_id`
- `staff_id → security_clearance`

BCNF Check: All FDs have staff_id as determinant, which is the sole candidate key.

✓ guard is in BCNF.

17. ROUTE_STATION

Attributes: route_id, station_id, visiting_order, arrival_time, departure_time, halt_time, distance_from_source

Candidate Keys: {route_id, station_id}

All FDs:

- `(route_id, station_id) → visiting_order`
- `(route_id, station_id) → arrival_time`
- `(route_id, station_id) → departure_time`
- `(route_id, station_id) → halt_time`
- `(route_id, station_id) → distance_from_source`

BCNF Check: All FDs have the composite PK {route_id, station_id} as determinant, which is the sole candidate key (and thus a superkey).

No partial or transitive dependencies exist.

✓ **route_station is in BCNF.**

18. SCHEDULE

Attributes: train_id, starting_ts, running_time, ending_ts, source_station_id, destination_station_id

Candidate Keys: {train_id, starting_ts}

All FDs:

- `(train_id, starting_ts) → running_time`
- `(train_id, starting_ts) → ending_ts`
- `(train_id, starting_ts) → source_station_id`
- `(train_id, starting_ts) → destination_station_id`

BCNF Check: All non-trivial FDs have the full key $(train_i d, starting_t s)$ on the left, so every determinant is a superkey and the SCHEDULE relation is in BCNF.

✓ **schedule is in BCNF.**

19. PAYMENT

Attributes: transaction_id, amount, method, payment_time

Candidate Keys: {transaction_id}

All FDs:

- `transaction_id → amount`
- `transaction_id → method`
- `transaction_id → payment_time`

BCNF Check: All FDs have transaction_id as determinant, which is the sole candidate key.

✓ **payment is in BCNF.**

20. TICKETS

Attributes: pnr, passenger_count, booking_ts, user_id, transaction_id, boarding_station, destination_station, schedule_train_id, schedule_starting_ts

Candidate Keys: {pnr}, {transaction_id}

All FDs:

- `pnr → passenger_count`
- `pnr → booking_ts`
- `pnr → user_id`
- `pnr → transaction_id`
- `pnr → boarding_station`
- `pnr → destination_station`
- `pnr → schedule_train_id`
- `pnr → schedule_starting_ts`
- `transaction_id → schedule_starting_ts, schedule_train_id, boarding_station, user_id, booking_ts, passenger_count, pnr`

BCNF Check: All determinants (pnr, transaction_id) are candidate keys, so every FD has a superkey on the left and the TICKETS relation is in BCNF.`

✓ tickets is in BCNF.

21. PASSENGER

Attributes: passenger_id, pnr, coach_code, seat_number, name, gender, dob, birth_pref

Candidate Keys: {passenger_id, pnr}

All FDs:

- `(passenger_id, pnr) → coach_code`
- `(passenger_id, pnr) → seat_number`
- `(passenger_id, pnr) → name`
- `(passenger_id, pnr) → gender`
- `(passenger_id, pnr) → dob`
- `(passenger_id, pnr) → birth_pref`

BCNF Check: In PASSENGER, the only determinant in all FDs is the composite key (passenger_id, pnr), which uniquely identifies each tuple.
Since every non-trivial FD has this candidate key (hence a superkey) on the left, the relation satisfies the BCNF condition.

✓ passenger is in BCNF.

22. COACH

Attributes: coach_code, train_id, coach_type, fare_multiplier, total_seats

Candidate Keys: {coach_code, train_id}

All FDs:

- `(coach_code, train_id) → coach_type`
- `(coach_code, train_id) → fare_multiplier`
- `(coach_code, train_id) → total_seats`

BCNF Check: All non-trivial FDs have the full key `(coach_code, train_id)` on the left, so every determinant is a superkey and the relation is in BCNF.

No partial or transitive dependencies.

✓ coach is in BCNF.

23. SEAT

Attributes: train_id, coach_code, seat_number, berth_type

Candidate Keys: {train_id, coach_code, seat_number}

All FDs:

- `(train_id, coach_code, seat_number) → berth_type`

BCNF Check: All non-trivial FDs in SEAT have the full key on the left, so the relation is in BCNF.

✓ seat is in BCNF.

24. TICKET_REFUND

Attributes: refund_id, pnr, reason_code, refund_amount, refund_date

Candidate Keys: {refund_id}

All FDs:

- `refund_id → pnr`
- `refund_id → reason_code`
- `refund_id → refund_amount`
- `refund_id → refund_date`

BCNF Check: All FDs have refund_id as determinant, which is the sole candidate key.

✓ ticket_refund is in BCNF.

25. LIVE_TRAIN_STATUS

Attributes: schedule_train_id, schedule_starting_ts, station_id, delay_minutes, delay_reason, reported_time, status

Candidate Keys: {schedule_train_id, schedule_starting_ts, station_id}

All FDs:

- `(schedule_train_id, schedule_starting_ts, station_id) → delay_minutes`
- `(schedule_train_id, schedule_starting_ts, station_id) → delay_reason`
- `(schedule_train_id, schedule_starting_ts, station_id) → reported_time`
- `(schedule_train_id, schedule_starting_ts, station_id) → status`

BCNF Check: All FDs have the composite PK {schedule_train_id, schedule_starting_ts, station_id} as determinant, which is the sole candidate key.

Partial dependency check:
• (schedule_train_id, schedule_starting_ts) → delay_minutes? No — delay is specific to each station on the journey.
• station_id → delay_minutes? No — different trains have different delays at the same station.

No partial or transitive dependencies.

✓ live_train_status is in BCNF.